



Ramachandran N

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📅 10/04/1993

👤 Carrer Synopsis

- **Technical Specialist who steered product transformation by reengineering existing process**; developing and executing strategic initiatives for long-range planning and lifecycle management for various products.
- Subject matter expertise in the emerging technologies in industry like **Robotics, IoT, AR & VR, ML, Cloud Computing**.
- **Expertise in setting a vision for how technology will be used in the company**; ensuring that technological resources meet the company's short and long-term needs.
- Excellence in formulating & implementing Agile transformation & innovation strategies; managing Agile Transformation Programs & helping teams with Agile adoption & implementation.
- Proficient in devising strategies to **automate the processes** by utilizing technologies; ***designed and developed multiple automated solutions and addressed in Smart India Hackathons, KPIT SPARKLE, Techgium and International events.***
- Merit of leading research initiatives, securing grants, and publishing scholarly articles in reputable journals, contributing to the advancement of knowledge in the field; ***successfully published 23 Papers, 13 Patents and 3 Books.***
- Recognized for spearheading the creation of the **Society of Automation and Robotics Club** and designed the club's goal, vision, and objectives aligning them with the wants and needs of students interested in automation and robotics.
- Extensive experience in **defining project scope and schedule while focusing on regular and timely delivery** of value, organizing and leading project status.
- **Efficient organizer, motivator, team player and a decisive leader** with the capability to motivate teams to excel.

Education

Pursuing Ph.D., <i>Sri Krishna College of Engineering and Technology</i>	Present Coimbatore, Tamil Nadu
ME, Bannari Amman Institute of Technology, Anna University	2016 Coimbatore, Tamil Nadu
BE, Shri Angalamman College of Engineering & Technology, Anna University	2014 Trichy, Tamil Nadu

Core Competencies

Technical Excellence	● ● ● ● ●	Big Data Analytics & Cloud Computing	● ● ● ● ●
AI, ML, AR, VR Technologies	● ● ● ● ●	Robotics, IoT, Embedded Systems	● ● ● ● ●
Client & Stakeholder Engagement	● ● ● ● ●	Industrial Automation	● ● ● ● ●
Project Management & Execution	● ● ● ● ●	Continuous Improvement	● ● ● ● ●

Soft Skills

Leadership	=====	Collaborator	=====
Critical Thinker	=====	Planner	=====
Detail Oriented	=====	Decision Maker	=====

Professional Experience

Assistant Professor, <i>Sri Krishna College of Engineering and Technology</i>	2016 – present Coimbatore
CEO & Founder, NAKAZA AUTOMATIONS <i>An Education Based Start-Up</i>	

Key Result Areas

- Establishing the business processes to ensure process optimization, quality checks and process adherence thereby delivering cost benefits
- Steering the deployment of the latest technologies and required at IT processes.
- Managing risk, impact, and guidelines through engagement with all stakeholders.

- Leading technical innovation and automation for the services provided to customers.
- Organizing technical teams, distributing work and monitoring progress, quality assurance, and performance in a multidisciplinary team environment.
- Developing the strategy for using technological resources while ensuring technologies are used efficiently, profitably and securely.
- Developing technical aspects of the strategy to ensure alignment with its business goals.

Awards & Accolades

- Grants Received Award
- Patent Published Award
- Star Performer Award
- Young Achieving Award

Grants Received

Grants Received Award

25 lakhs, from SYINDIA Pvt. Ltd.

Fund, from TNSCST (Tamil Nadu State Council for Science and Technology)

3 lakhs, from Ministry of AYUSH

4 lakhs, from AICTE through YUKTI Innovation Challenge 2023

Positions of Responsibility

- Smart India Hackathon (SIH) College Coordinator
- Society of Automation and Robotics (SAR) Club In-charge
- Society of Automotive Engineers (SAE-INDIA) Faculty Member
- Jury among top 20 for National Hackevent - Samadhan-AICTE-MHRD
- AICRA Coordinator of SKCET
- Department Faculty Project Coordinator Team Head
- College all File Maintenance In-charge
- Department Activity Saturday Coordinator In-charge
- NAAC Criteria 5 File In-charge
- Department Budget File In-charge
- HOR Men Residential Warden
- Report Analyst at SKCET Institution

Projects Undertaken

- Plug-In Kit for Ready Charging
- AI Based Smart Yoga Mat
- AR VR Based Virtual Herbarium
- IoT Based Early Detection of Muscle Cramp/Spasms
- AI Based Kashaya Making Machine
- AR VR Based Students Laboratory Application
- ML Incorporated Smart Irrigation
- Virtual Reality Based Virtual Yoga Tutor
- NAKAZA – Automated Water Hyacinth Removal Rover
- Smart Dam Monitoring System
- Smart Earthing Leakage Detection System
- Microbial Detection in Drinking Water without contamination
- Myoelectric Prosthetic Human Arm
- Smart AI Writing Pen and Pad for Physically Challenged
- Automatic Firefighting Robot with Smart Camera
- Design and Fabrication of Stair Climbing Load Carrier
- Sensor Networking and Embedded Technology Applied Dirt Bikes

Patents Published

Smart Surveillance Energy Meter Using Iot (Application Number - 202341004657)

Iot Based Logistic Maintenance System (Application Number - 202341003710)

Fuel Injector Relative Performance Tester And Cleaner (Application Number - 202241075960)

IOT based Child Rescue System from an Open Borewell Unit (Application Number -202241024020)

AI based Ambulant Toilets (Application Number - 202241010466)

RFID based bus ticketing system (Application Number - 202041039793)

Brake and Application using Single Pedal (Application Number - 202041039928)

Intelligent Wheelchair with Voice User Interface System for Disabled and Elderly using IOT (Application Number - 202041007185)

Multistage Telescopic Portable Seater (Application Number - 202041008129)

Air Engine Powered Vehicle for Transportation (Application Number - 202041000112)

Automatic Fire Fighting Robot with Smart Camera (Application Number - 201941051826)

IOT Based Detection of Level of Pesticide in Fruits and Vegetables (Application Number - 201941009173)

Design and Fabrication of Stair Climbing Load Carrier (Application Number - 201941009167)

Papers Published

- Ramachandran N et.al, Review On Optical Tweezers In Multifields'AIP Conference Proceedings (Accepted).
- Ramachandran N et.al, Flexural Behaviour Of Coconut Inflorescence Fibre Reinforced Epoxy Composites'AIP Conference Proceedings(Accepted).
- Ramachandran N et.al, 'The Experimental Development and Enhancement of Ultrasonic Welding Process Specification Aimed at Delrin' AIP Conference Proceedings (Accepted).
- Ramachandran N et.al, 'Assessment Of Mechanical Properties OnTIG Welded Dissimilar Metal Welded Joints' Materials Today Proceedings August 2022.
- Ramachandran N et.al, 'Design and Fabrication of Hybrid Mobility Scooter',Part of the Springer Proceedings in Materials book series (SPM, volume 5), Advances in Materials Research pp-645-654 February 2021.

- Ramachandran N et.al, Design and Fabrication of Steam Sanitiser' ,International Journal of Engineering and Advanced Technology (IJEAT), ISSN: 2249-8958(Online), Volume-9 Issue-3 February 2020.
- Ramachandran N et.al, Design and Fabrication of Beach Cleaning Machine', International Journal of Innovative Technology and Exploring Engineering, ISSN: 2278-3075, Volume-8 Issue-12, October 2019.
- Ramachandran N et.al, 'Effect on the Mechanical Properties of Za-27/Graphite Reinforced Composites when Routed through a Squeeze Casting Process', International Journal of Innovative Technology and Exploring Engineering, ISSN: 2278-3075, Volume-8 Issue-10, August 2019.
- Ramachandran N et.al, 'Design and fabrication of Semi-Automated Lawn Mower' , International Journal of Innovative Technology and Exploring Engineering, ISSN: 2278-3075, Volume-8 Issue-10, August 2019.
- Ramachandran N et.al, Design And Fabrication Of Stair Climbing Load Carrier' , International Journal of Innovative Technology and Exploring Engineering, ISSN: 2278-3075, Volume-8 Issue-4, February 2019.
- Ramachandran N et.alatarajan, 'Valediction Of Residual Stress Measured Using Hole- Drilling Method With 3d Simulation That Developed During TIG Welding Of AISI 304 Ss Plates', International Journal of Mechanical Engineering and Technology (IJMET), Volume 10, Issue 01, January 2019, pp. 867–873, Article ID: IJMET_10_01_090,2019
- Ramachandran N et.alatarajan, Design And Development Of Weight Based Material Sorting And Feeding System' International Journal of Pure and Applied Mathematics, Volume 119 No. 12 2018, 2101-2108 ISSN: 1314-3395 (on-line version) url: <http://www.ijpam.eu> Special Issue.
- Ramachandran N et.al, 'Experimental Investigation On Wire-Electro Discharge Machining Of Tungsten Carbide (Wc) Using RSM' International Journal of Mechanical Engineering and Technology (IJMET), Volume 9, Issue 13, December 2018, pp. 1101–1108, IJMET_09_13_115,2018.
- Ramachandran N et.al, 'Methods For Optimal Design And Enhancement of Performance of Heat Pipe – A Review' International Journal for Science and Advance Research in Technology, Volume 3, issue 11, page 59-64, November 2017.
- Ramachandran N et.al, 'Design and Fabrication of Brake and Accelerator by Using Single Pedal', International Journal of Advance Research in Science and Engineering volume 6, Issue no.7, pages 276- 283.
- Ramachandran N et.al, 'Experimental Investigation on Solar Water Heater using Scheffler Reflector' International Journal of Advanced research in Science and Engineering, volume 6, Issue no.7, pages 538-553. July 2017.
- Ramachandran N et.al, Experimental Investigation And Optimization Of Process Parameters Of Ultrasonic Welding For Delrin Material' International Journal of Advance Research in Science and Engineering, Volume 6, Issue no.4, pages 883-890, April 2017.

- Ramachandran N et.al, 'Investigation on Combustion and Emission Analysis in a CI Engine Using CFD', International Journal of Applied Engineering and Research, Volume 10, Special issue, page no.252-260,2015.
- Ramachandran N et.al, 'Experimental Investigation and Performance Improvement of Modified Solar Still' International Journal of Science and Advance Research in Technology, Volume 3 Issue 11 – NOVEMBER 2017, page no. 169-177, November 2017.
- Ramachandran N et.al, 'Enhancement of Thermal Conductivity Using Nanofluids' International Journal of Science and Advance Research in Technology, Volume 3 Issue 11 – NOVEMBER 2017, page no. 144-153, November 2017.
- Ramachandran N et.al, Design and Fabrication of 90 Degree Car Parking System' International Journal of Applied Engineering and Research (IJAER) Vol 10, Number 50 (2015) Special Issues pp 342 – 352.
- Ramachandran N et.al, 'Enhancement of Conventional Baby Diaper Making Machine Process Using Programmable Logic Controller' International Journal of Science and Advanced Research in Technology, Issn No: 0090-5674 Volume 13 Issue I 2020.
- Ramachandran N et.al, 'Development and Conception of Versatile Agricultural Machine' International Journal of Science and Advanced Research in Technology, Proceedings 37 (2021) 2582–2586.

Book Chapter Published

- Handbook of Research on Intelligent Industrial IoT: "Compelling Forces and Challenges for a Food Processing Industry to adopt Industry 4.0".
- "Technology Advances and New Applications by Apple Academic Press" on the book named Internet of Things
- "Influence of Coating on Reinforcements on Mechanical Properties and Wear Behaviour of Metal Matrix Composites -A Review" on the book, Emerging Trends in Mechanical Computing and Electrical Science (ISBN:978-81-941281-8-2)

Events Won

Plug-in-Kit for Ready Charging (Won Gold Award with prize of Rs 5 Lakh) – **KPIT SPARKLE'24**

Organised by: KPIT Technology Private Limited

HYDRA – Smart Dam Monitoring System (Won first prize of 1 Lakh) – **Smart India Hackathon – Hardware (SIH 2023)**

Organised by: AICTE-MHRD

NAKAZA – Automated Water Hyacinth Removal Rover (Won first prize of 1 Lakh) – **Smart India Hackathon – Hardware (SIH 2023)**

Organised by: AICTE-MHRD

Plug-in Kit for Ready Charging (Won First Runner prize of worth 5 lakh) - **L&T Techgium'23**

Organized by: LTTS, Mysore

IoT Based Early Detection of Muscle Cramp/Spasms (Won Employee Choice Award) - **L&T Techgium'23**

Organized by: LTTS, Mysore

AI Based Kashaya Making Machine (Won first prize of 1 lakh) - **Smart India Hackathon - Hardware (SIH 2022)**

Organized by: AICTE-MHRD

AR VR-Based Student's Laboratory Application (Won First prize of worth 1 lakh) - **Smart India Hackathon - Hardware (SIH 2022)**

Organized by: AICTE-MHRD

Meta Lab (Won First Runner up prize of worth 300€) - **EDXR Challenge-2022**

Organized by: University College Dublin

Decentralized Application for Freelancing Platform using XDC Network (Won prize of worth Rs.25,000 in XDC coins) - **PLI Blockathon Inida-2022**

Organized by: University College Dublin

o Design and Fabrication of an IoT Based Digital Clock –2022

Design and Fabrication of Model Rocket- APJ-AK-01

India Design Week-2022 (1st Prize)

Organized by: AutoDesk & ICT Academy

RUT Riders (Rs. 10,000) - **MEQUEST'24-Dirt Bike Race**

Organized by: Sri Ramakrishna Engineering College, Coimbatore

Rage Mantels (Rs. 3,000) - **MEQUEST'24-Dirt Bike Race**

Organized by: Sri Ramakrishna Engineering College, Coimbatore

Rage Mantle (Rs. 10,000) – **RIDE'24 - National Level Dirt Bike Race**

Organized by: Karpagam Institute of Technology, Coimbatore

Rage Mantle (Rs. 10,000) – **RIVALS'24 - National Level Dirt Bike Race**

Organized by: Karpagam College of Engineering, Coimbatore

Rebels (Rs. 5,000) - **RIVALS'24 - National Level Dirt Bike Race**

Organized by: Karpagam College of Engineering, Coimbatore

Rebels (Rs. 10,000) - **MEQUEST'23-Dirt Bike Race**

Organized by: Sri Ramakrishna Engineering College, Coimbatore

Rebels (Rs. 10,000) - **RPM 2k24- National Level Dirt Bike Race**

Organized by: SNS Institute of Technology, Coimbatore

RUT Riders (Rs. 4,000) - **RPM 2k23- National Level Dirt Bike Race**

Organized by: SNS Institute of Technology, Coimbatore

RUT Riders (Rs. 3,000) - **MEQUEST'23-Dirt Bike Race**

Organized by: Sri Ramakrishna Engineering College, Coimbatore

Bettmenn's Crew (Rs. 15,000)

Organized by: Karpagam College of Engineering, Coimbatore

Rage Mantle (Rs. 10,000) - **RPM 2k23- National Level Dirt Bike Race**

Organized by: SNS Institute of Technology, Coimbatore

Rebels (Rs. 10,000) - **RALLERZ'23- National Level Dirt Bike Race**

Organized by: Sri Ramakrishna Institute of Technology, Coimbatore

Rebels (Rs. 5,000) - **MEQUEST'22-Dirt Bike Race**

Organized by: Sri Ramakrishna Engineering College, Coimbatore

Rebels (Rs. 6,000) - **RPM 2k23- National Level Dirt Bike Race**

Organized by: SNS Institute of Technology, Coimbatore

RUT Riders (Rs. 4,000) - **RPM 2k23- National Level Dirt Bike Race**

Organized by: SNS Institute of Technology, Coimbatore

RUT Riders (Rs. 15,000) - **RIVILS'23 Dirt Bike Race**

Organized by: Karpagam College of Engineering, Coimbatore

RUT Riders (Rs. 5,000) - **RALLERZ'23- National Level Dirt Bike Race**

Organized by: Sri Ramakrishna Institute of Technology, Coimbatore

Virtual Reality Based Yoga-Tutor (Rs. 1 lakh) - **Smart India Hackathon - Hardware (SIH 2020)**

Organized by: AICTE-MHRD

Crowd Control Monitoring Application with Advanced Mask and Sanitiser Inbuilt Watch (USD 200 + e-Certificate) - **International IoT Challenge, Indonesia- 2021**

Organized by:Sebelas Maret University, Indonesia

Augumented Reality Based Game of Cards (Selected for Grand Finale) -

TOYCATHON'21-AICTE-MHRD – 2021

Organized by: AICTE-MHRD

Automatic Firefighting Robot with Smart Camera (Rs.5000 + Certificates) - **Tech**

QUEST'19, India– Caterpillar Manufacturing Day-2019

Organized by: Caterpillar India PVT LTD,Chennai.

Artificial Intelligence (AI) Based Personal Protective Equipment (11th Place) - **MIT –**

COVID'19 Challenge (MIT), USA- 2019

Organized by: Massachusetts Institute of Technology (MIT),USA

Assisting Equipment for Ventilator (29th Place) - **MIT –COVID'19 Challenge (MIT),**

USA-2019

Organized by: Massachusetts Institute of Technology (MIT),USA.

AI Based Contraption (Rs. 10000; Innovation Award) - **KovaiVizha ' 19-2019**

Organized by: Codissia,Coimbatore

Last Mile Delivery –Drone (Rs. 7500; 2nd Prize) - **VISAI 2020 -2020**

Organized by: Veltech University,Chennai

Yeshe- The Solar Bike (1st Prize) - **KRIYA2020-Paper Presentation-2020**

Organized by:PSG College Of Technology, Coimbatore

Face Mask, Shields and Ventilator Splitter (Milestone 2 Round) - **IDEATHON –**

COVID'19 Challenge-2019

Organized by: MHRD's Innovation Cell, AICTE.

HoverBoat Contest, Hove Rush (Rs. 10000; 2nd Prize) - **AAKRUUSH'19" -2019**

Organized by: SRM Institute of Technology and Sciences,Chennai

Ultimate Hands-on (2nd Prize) - **MECHANICA2k18 – 2018**

Organized by: IIT Madras

Project X (2nd Prize) - **MECHANICA2k18 – 2018**

Organized by: IIT Madras

Automatic Lawn Mower (1st Prize) - **Tech Fest- Project Contest- 2018**

Organized by: Karunya Institute Of Technology

Title-Automatic Paper Pouch Making Machine (1st Prize) - **ORACLE 2K18- Project Contest-2018**

Organized by: Government College Of Technology, Coimbatore

Unconference E-summit -2018 (3rd Prize)

Organized by: IIT Madras

Wall Climbing Robot (1st Prize) - **ORACLE 2K18-Project Contest-2018**

Organized by: Hindustan College of Engineering and Technology, Coimbatore

Automatic Paper Pouch Making Machine (1st Prize) - **Futura'17 -Project Contest-2017**

Organized by: Bannari Amman Institute of Technology, Sathy

Automatic Lawn Mower (1st Prize) - **NOVA' 17 - Project Contest-2017**

Organized by: CIT, Coimbatore

RFID Based Student Healthcare Application – **DakshHackathon**

Organized by: SASTRA Deemed to be University, Tanjore

YESHO- The Solar Bike (1st Prize) - **Dhruva-Paper Presentation-2019**

Organized by: Karpagam College Of Engineering, Coimbatore

Jury Participation

- Judge at the '**Veetuku Oru Vignani**' event, hosted by Puthiyathalaimurai TV Television
- Have been an idea evaluator for a set of Problem Statements under Smart India Hackathon 2023, presented by AICTE
- Juried a national level Hackathon from the Department of Computer Science Engineering held at Sri Krishna College of Engineering and Technology
- Juried a coding contest event conducted by the Department of Computer Science and Business Systems at SKCET, Coimbatore

Guest Lectures Delivered

Innovation/Prototype Validation-Converting Innovation into a Start-up — Institution Innovation Council

Prototype Process Design and Development-Prototyping — Institution Innovation Council

Engineers View on Engineering — Dhanalakshmi Srinivasan Engineering College

Scope of IoT in the Mechanical World — Indra Ganesan College of Engineering

Internet of Things — Bannari Amman Institute of Technology

Internet of Things for Mechanical Engineers — Jeppiaar Institute of Technology

Languages

Tamil — Native/Bilingual

English — Fluent